```
!pip install spacy
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.5)
Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.0.15)
Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.13)
Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (3.0.12)
Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.12/dist-packages (from spacy) (8.3.10)
Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.12/dist-packages (from spacy) (1.1.3)
Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.5.2)
Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.10)
Requirement already satisfied: weasel<0.5.0,>=0.4.2 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.4.3)
Requirement already satisfied: typer-slim<1.0.0,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (0.24.0)
Requirement already satisfied: tqdm<5.0.0,>=4.38.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (4.67.3)
Requirement already satisfied: numpy>=1.19.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.0.2)
Requirement already satisfied: requests<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from spacy) (2.32.4)
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Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0
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Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1.0.0,>=0
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer>=0.24.0->typer-slim<1
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.24.
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer>=0.2
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Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any
     ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 12.8/12.8 MB 79.5 MB/s eta 0:00:00
✓ Download and installation successful
You can now load the package via spacy.load('en_core_web_sm')
⚠ Restart to reload dependencies
If you are in a Jupyter or Colab notebook, you may need to restart Python in
order to load all the package's dependencies. You can do this by selecting the
'Restart kernel' or 'Restart runtime' option.
```

```
import spacy

nlp = spacy.load("en_core_web_sm")
print("Model loaded successfully!")
```

```
Model loaded successfully!
```

```
sentences = [
    "Apple is looking at buying U.K. startup for $1 billion.",
    "Google is launching a new AI assistant this week.",
    "Tesla announced record profits last quarter."
]

docs = [nlp(sentence) for sentence in sentences]

for doc in docs:
    print(f"\nSentence: {doc.text}")
    for token in doc:
        print(token.text, token.pos_, token.dep_)
```

```

Sentence: Apple is looking at buying U.K. startup for $1 billion.
Apple PROPN nsubj
is AUX aux
```

```
looking VERB ROOT
at ADP prep
buying VERB pcomp
U.K. PROPN nsubj
startup VERB ccomp
for ADP prep
$ SYM quantmod
1 NUM compound
billion NUM pobj
. PUNCT punct

Sentence: Google is launching a new AI assistant this week.
Google PROPN nsubj
is AUX aux
launching VERB ROOT
a DET det
new ADJ amod
AI PROPN compound
assistant NOUN dobj
this DET det
week NOUN npadvmod
. PUNCT punct

Sentence: Tesla announced record profits last quarter.
Tesla PROPN nsubj
announced VERB ROOT
record NOUN compound
profits NOUN dobj
last ADJ amod
quarter NOUN npadvmod
. PUNCT punct
```

```
import pandas as pd

data = []

for doc in docs:
    for token in doc:
        data.append([
            doc.text,
            token.text,
            token.lemma_,
            token.pos_,
            token.dep_,
            token.head.text
        ])

df = pd.DataFrame(data, columns=[
    "Sentence", "Token", "Lemma", "POS", "Dependency", "Head"
])

print(df.head(20))
```

```
                                            Sentence      Token      Lemma  \
0   Apple is looking at buying U.K. startup for $1...      Apple      Apple
1   Apple is looking at buying U.K. startup for $1...         is         be
2   Apple is looking at buying U.K. startup for $1...    looking       look
3   Apple is looking at buying U.K. startup for $1...         at         at
4   Apple is looking at buying U.K. startup for $1...     buying        buy
5   Apple is looking at buying U.K. startup for $1...       U.K.       U.K.
6   Apple is looking at buying U.K. startup for $1...    startup    startup
7   Apple is looking at buying U.K. startup for $1...        for        for
8   Apple is looking at buying U.K. startup for $1...          $          $
9   Apple is looking at buying U.K. startup for $1...          1          1
10  Apple is looking at buying U.K. startup for $1...    billion    billion
11  Apple is looking at buying U.K. startup for $1...          .          .
12  Google is launching a new AI assistant this week.     Google     Google
13  Google is launching a new AI assistant this week.         is         be
14  Google is launching a new AI assistant this week.  launching     launch
15  Google is launching a new AI assistant this week.          a          a
16  Google is launching a new AI assistant this week.        new        new
17  Google is launching a new AI assistant this week.         AI         AI
18  Google is launching a new AI assistant this week.  assistant  assistant
19  Google is launching a new AI assistant this week.       this       this

      POS Dependency       Head
0   PROPN      nsubj    looking
1     AUX        aux    looking
2    VERB       ROOT    looking
3     ADP       prep    looking
4    VERB      pcomp         at
5   PROPN      nsubj    startup
```

```
6    VERB      ccomp     buying
7     ADP       prep    startup
8     SYM   quantmod    billion
9     NUM   compound    billion
10    NUM       pobj        for
11  PUNCT      punct    looking
12  PROPN      nsubj  launching
13    AUX        aux  launching
14   VERB       ROOT  launching
15    DET        det  assistant
16    ADJ       amod  assistant
17  PROPN   compound  assistant
18   NOUN       dobj  launching
19    DET        det       week
```

```
for doc in docs:
    print(f"\nSentence: {doc.text}")
    subject = [tok.text for tok in doc if tok.dep_ == "nsubj"]
    obj = [tok.text for tok in doc if tok.dep_ == "dobj"]
    root = [tok.text for tok in doc if tok.dep_ == "ROOT"]

    print("Subject:", subject)
    print("Verb (ROOT):", root)
    print("Object:", obj)
```

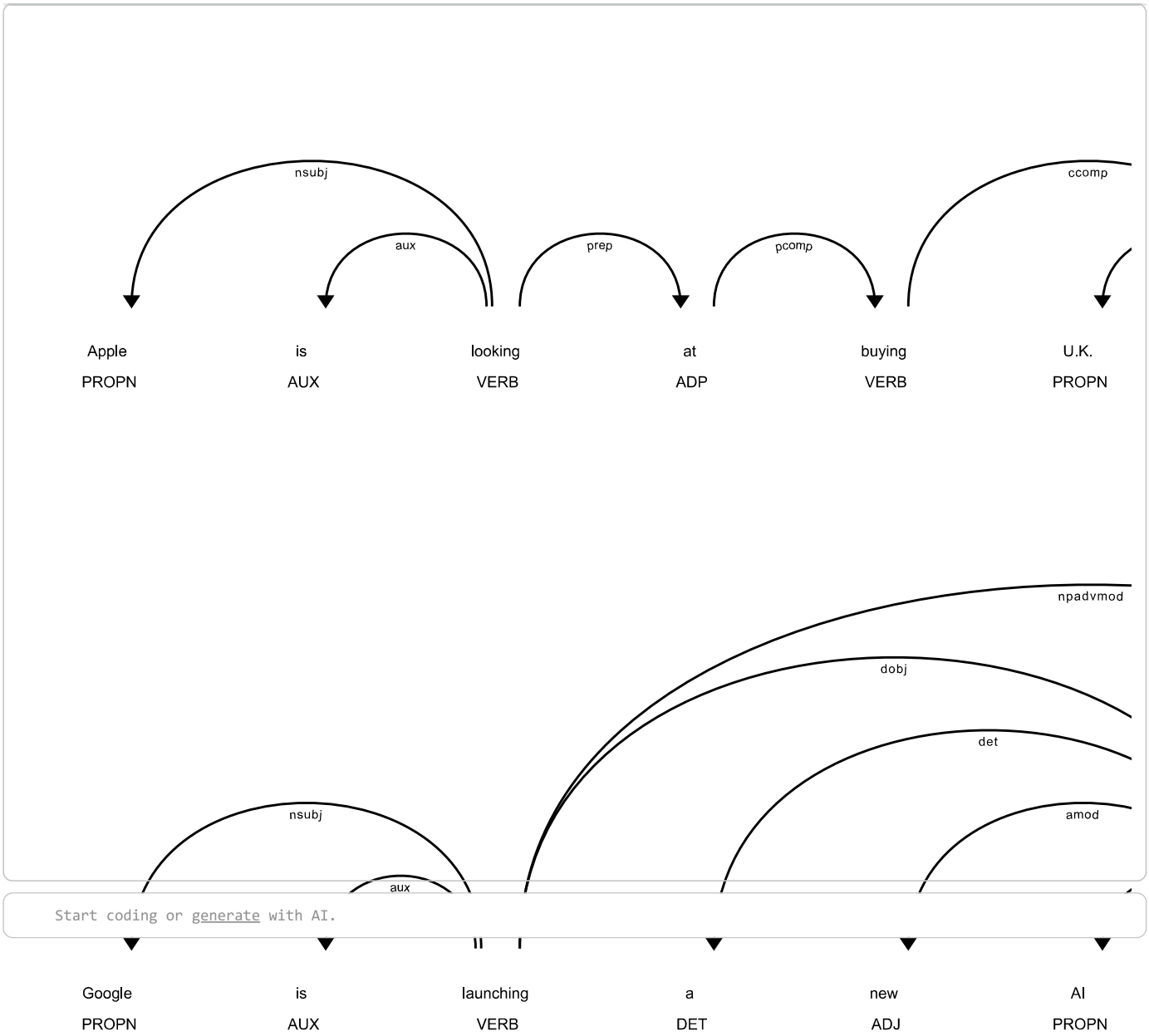
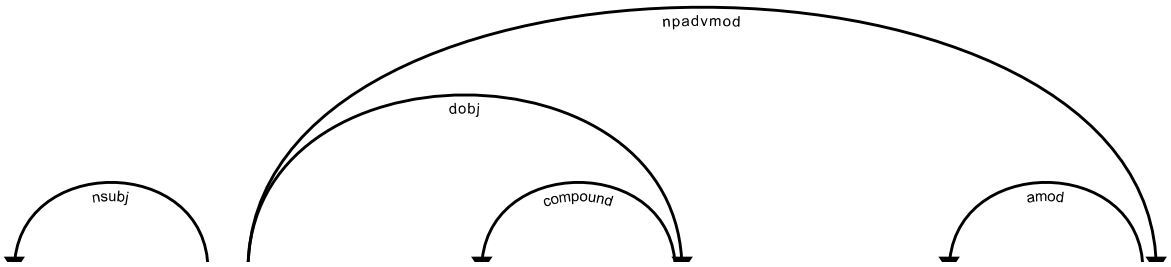
```
Sentence: Apple is looking at buying U.K. startup for $1 billion.
Subject: ['Apple', 'U.K.']
Verb (ROOT): ['looking']
Object: []

Sentence: Google is launching a new AI assistant this week.
Subject: ['Google']
Verb (ROOT): ['launching']
Object: ['assistant']

Sentence: Tesla announced record profits last quarter.
Subject: ['Tesla']
Verb (ROOT): ['announced']
Object: ['profits']
```

```
from spacy import displacy

for doc in docs[:5]:
    displacy.render(doc, style="dep", jupyter=True)
```



Start coding or generate with AI.